

DL for ENG

## LECTURE 9.2

# Gas Flow

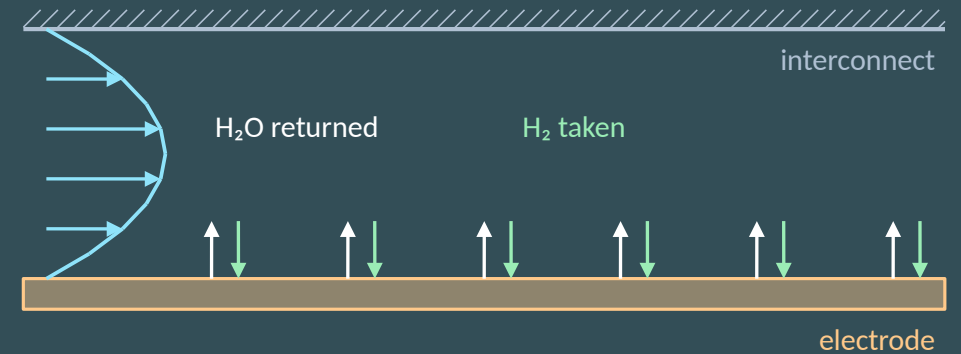
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### GAS · SPECIES · EVERY SPEED



*The fuel channel of a solid oxide cell. Three facts make its gas behave like the liquid of the last lecture, so the flow is known; the question is what the gas carries - hydrogen in, steam out - and one PINN, with the speed as an input, answers it for every speed and finds the speed a design asks for.*

# Learning Objectives

## SECTIONS, IN ORDER

- 1 **The gas** — slow, one for one, Poiseuille from the inlet
- 2 **What the gas carries** — the species equation, its solution
- 3 **Every speed** — one PINN, and the speed a design needs

*White carries the gas, Incropera the mass transfer and Liu the method; the example is Ex\_09.2's fuel channel; L10.2 adds the electrode behind it.*

## READ, AND WHAT EACH ANSWERS

**White ch. 9**      when is a gas incompressible?

**Incropera ch. 14**      how is a species carried?

**Liu ch. 2.2**      what does an input speed buy?

# The Fuel Channel

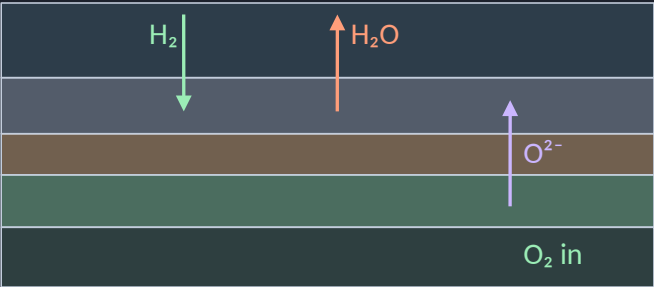
**Fuel channel** — the gap between a solid oxide cell's fuel electrode and its interconnect. Oxide ions crossing from the air side burn the hydrogen in the porous electrode, and the steam made returns to the gas: one molecule each way for every two electrons.

## THE REACTION, AND THE FLUX IT SETS

$$N = \frac{i}{2F} \qquad \text{H}_2 + \text{O}^{2-} \rightarrow \text{H}_2\text{O} + 2\text{e}^-$$

$N$  molecules exchanged per area and second,  $i$  the current density,  $F$  Faraday's constant

BEHIND THE WALL: THE FUEL CELL



fuel channel  
fuel electrode, porous  
electrolyte: O<sup>2-</sup> only  
air electrode  
air channel

## THE CHANNEL OF EX\_09.2

|               | value                                                                   |
|---------------|-------------------------------------------------------------------------|
| channel       | 100 mm long, 1 mm high                                                  |
| gas in        | 97 % H <sub>2</sub> , 3 % H <sub>2</sub> O                              |
| state         | 800 °C, 1 atm: 11.36 mol/m <sup>3</sup>                                 |
| current       | 0.5 A/cm <sup>2</sup> : $N = 0.0259 \text{ mol}/(\text{m}^2 \text{ s})$ |
| inlet speed   | 0.3 to 1.5 m/s                                                          |
| run backwards | SOEC: 90 % steam fed, flux reversed                                     |

*Seen from the gas only; L10.2 adds the electrode. Driven by a supply the same cell splits steam and returns hydrogen: for the channel only the flux's sign changes (MP9.2A). The diffusivity, 8.0e-04 m<sup>2</sup>/s, is an estimate.*

# Gas That Flows Like a Liquid

**Incompressible, equimolar** — two facts make this gas flow like L9.1's liquid. Far below the speed of sound ( $Ma < 0.3$ ) the density ignores the pressure that drives the flow. A wall reaction returning one molecule for each it takes leaves the moles per volume at  $p/RT$ : the velocity is untouched.

## THE MACH NUMBER, AND AN IDEAL GAS

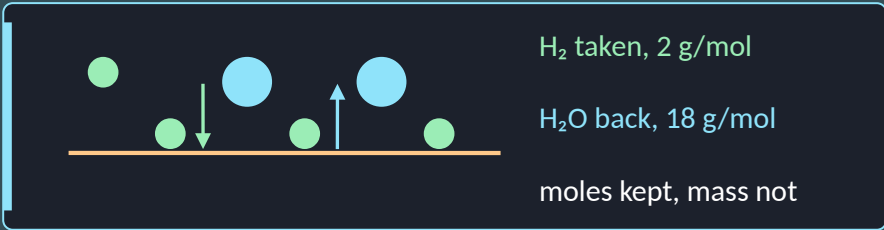
$$Ma = \frac{u}{a}, \quad a = \sqrt{\frac{\gamma RT}{M}}, \quad \frac{\Delta p}{\rho} \approx \frac{Ma^2}{2}, \quad c = \frac{p}{RT}$$

$a$  the speed of sound,  $\gamma$  the ratio of heat capacities,  $M$  the molar mass  
 $c$  moles per volume,  $p$  pressure,  $R$  the gas constant,  $T$  temperature

**TALK • 2 MIN** How would you explain, in your own words, why the reaction does not push the gas faster? Then, in the drawing on the right, what would change if two molecules entered for each one that left?

## AT 800 °C, AND ONE MOLECULE FOR ONE

| quantity             | value                                       |
|----------------------|---------------------------------------------|
| speed of sound $a$   | about 2240 m/s                              |
| Mach number          | below $10^{-3}$                             |
| density change       | about $10^{-7}$                             |
| moles per volume $c$ | 11.36 mol/m <sup>3</sup> , kept by the wall |



Light  $\text{H}_2$  leaves, heavy  $\text{H}_2\text{O}$  arrives: the mass rises along the channel, the moles and the velocity do not. So the species is counted in mole fractions.

# Poiseuille From the Inlet

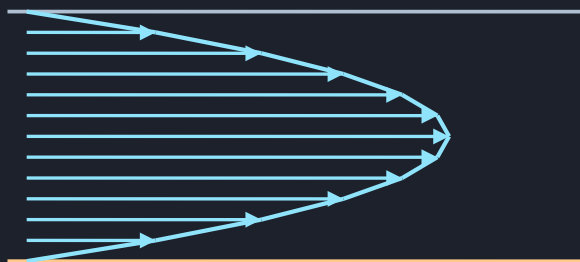
**Developed channel flow** — the flow between two plates far from the inlet: the parabola of L9.1 in a flat channel. Here the Reynolds number is 1.3 and the entrance length 0.06 mm, so the gas is fully developed from the inlet and its velocity is known everywhere before anything is solved.

## THE PARABOLA, AND ITS PRESSURE DROP

$$u(z) = 6\bar{u}\frac{z}{h}\left(1 - \frac{z}{h}\right), \quad \Delta p = \frac{12\mu\bar{u}L}{h^2}$$

$z$  across the channel,  $h$  its height,  $L$  its length,  $\bar{u}$  the mean speed,  $\mu$  viscosity

### ACROSS THE CHANNEL



zero on both walls

1.5 times the mean in the middle

formed within 0.06 mm

## THREE FACTS, THREE NUMBERS

| fact                    | because                  |
|-------------------------|--------------------------|
| incompressible          | Mach below $10^{-3}$     |
| velocity unchanged      | one molecule for one     |
| parabola from the inlet | Re 1.3, entrance 0.06 mm |

*Poiseuille's pressure drop over the channel is 26 Pa at 1 m/s. Darcy's law of L9.1 with  $K = h^2/12$ : the flow is solved, and the rest of the lecture is what it carries.*

# Convection–Diffusion

**Convection–diffusion** — the equation for a species carried by a known flow: the flow moves it along, diffusion spreads it from rich to poor. Its conditions are the inlet's composition, the flux the electrode returns, and an impermeable interconnect.

## THE STEAM FRACTION, AND ITS THREE CONDITIONS

$$u(z) y_x = D y_{zz}$$

$$y = y_{in} \text{ at } x = 0, \quad -Dc y_z = N \text{ at } z = 0, \quad y_z = 0 \text{ at } z = h$$

$y$  the steam fraction,  $x$  along,  $z$  across,  $D$  the diffusivity,  $N$  the wall flux

### THE TWO WAYS A SPECIES MOVES



along: carried by the flow

across: spread by diffusion

fed at the electrode

## ALONG AGAINST ACROSS

$$Pe_L = \frac{\bar{u}L}{D} = 125$$

**along** carried, Péclet 125

**across** spread, Péclet 1.25

**dropped** diffusion along:  $10^{-4}$

*The Péclet number compares carrying with spreading. Along the channel the flow wins by a hundred and twenty-five; across its height diffusion keeps up.*

# Marching Along the Channel

**Parabolic equation** — an equation with a direction in which information travels one way only. Without diffusion along it, the species equation is parabolic in  $x$ : it marches from the inlet as the heat equation of L8.2 marches in time, and it needs no condition at the outlet.

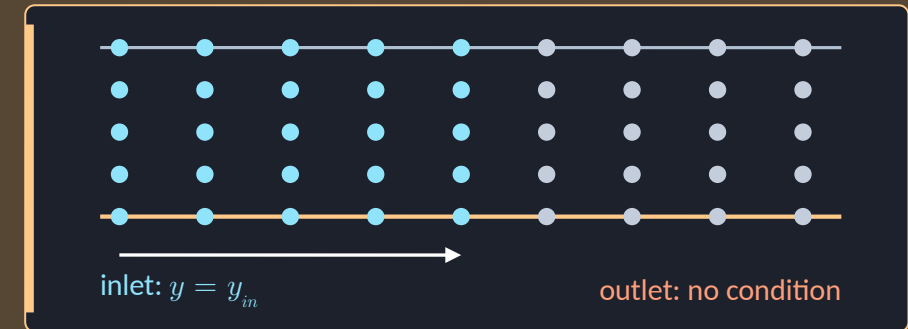
## THE SCALED EQUATION, MARCHED

$$6Z(1 - Z) G s \theta_X = \theta_{ZZ}, \quad G = \frac{u_{ref} h^2}{DL}$$

$X = x/L$ ,  $Z = z/h$ ,  $s = \bar{u}/u_{ref}$ ,  $\theta$  the rise of  $y$  in units of the channel's rise at  $u_{ref}$

**TALK • 2 MIN** How would you decide, in your own words, which conditions an equation needs and where? Then, in the drawing on the right, why would a zero-slope condition at the outlet be wrong for this channel?

## ONE COLUMN AT A TIME



FDM: Crank–Nicolson along, two implicit steps first, central differences across;  $401 \times 41$  nodes meet 0.1 percentage point at every speed in a few ms. The mean keeps rising to the last millimetre, so a zero-slope outlet would be wrong.

# Fully Developed Concentration

**Fully developed concentration** — the state beyond a short entrance in which the profile across the channel keeps its shape while the whole of it rises along. The rise of the mean is a balance: what the wall returns, the flow carries away.

## THE BALANCE, AND THE EXACT FIELD

$$\bar{y}(x) = y_{in} + \frac{Nx}{uhc}$$

$$\theta = \frac{x}{s} + G\varphi(Z), \quad \varphi = Z^3 - \frac{Z^4}{2} - Z + \frac{13}{35}$$

$\bar{y}$  the mixing cup mean,  $X/s$  that balance,  $\varphi$  the profile across, zero mean

ACROSS THE CHANNEL, AT ANY  $x$

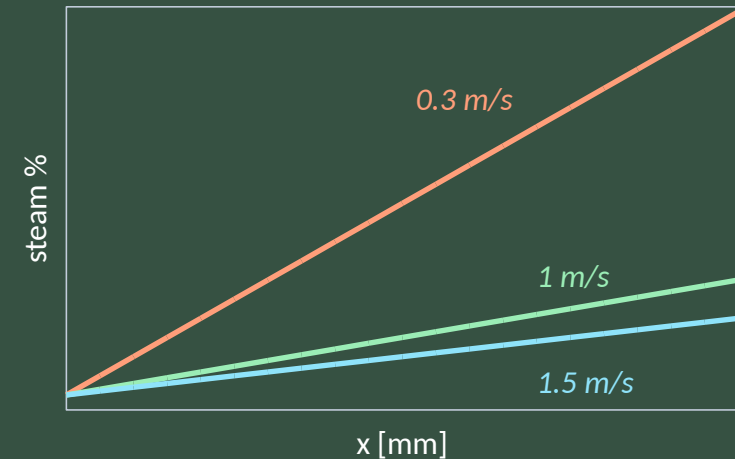


interconnect: a little less

dashed: the mean

electrode:  $13G/35$  above it

## STEAM ALONG THE CHANNEL



Ex\_09.2's exact field: from 3 % to 26 % at 1 m/s, to 79 % at 0.3 m/s. The electrode sees 0.11 percentage points more than the mean.

# The Sherwood Number

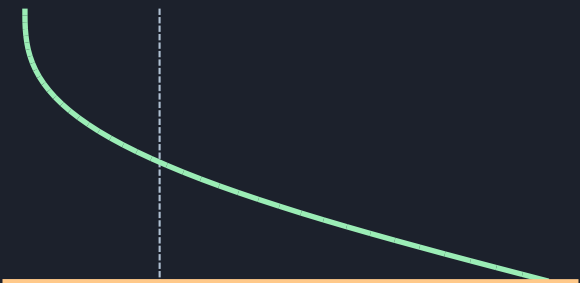
**Sherwood number** — the mass-transfer coefficient between a wall and the mean of the gas, made dimensionless with the channel's size and the diffusivity. It is the Nusselt number of heat transfer for a species: in a fully developed channel it is a constant of the geometry.

## THE WALL AGAINST THE MEAN

$$Sh = \frac{h_m 2h}{D} = \frac{2 \cdot 35}{13} = 5.385$$

$h_m$  the mass transfer coefficient,  $2h$  the hydraulic diameter,  $D$  the diffusivity

### THE GAP IT MEASURES



interconnect: a little less

dashed: the mean

electrode: 13G/35 above it

## WHAT EACH METHOD FINDS

|                   | Sherwood      |
|-------------------|---------------|
| exact             | 5.385         |
| FDM, 401 × 41     | 3.71 to 4.94  |
| PINN              | 6.12 to 6.75  |
| heat, same plates | Nusselt 5.385 |

One wall at a constant flux, the other impermeable: Shah & London's constant, and Incropera's table 8.1 for heat. Both methods find the field to a ten-thousandth but the Sherwood number only to 8 to 31 %: it is a tenth of a per cent of the field.

# One PINN for Every Flow Rate

**Parametric PINN** — a PINN that takes a parameter of the problem as an input beside the coordinates, here the inlet speed  $s$ . It is trained once over a range of speeds, and after that any speed is read off, not solved again: the one thing a march cannot do.

## THE BALANCE BUILT IN, THE PROFILE LEARNT

$$\hat{\theta}(X, Z, s) = \frac{X}{s} + G \mathcal{N}(X, Z, s)$$

$X/s$  the balance, built in,  $\mathcal{N}$  the network: the profile in units of  $G$

### THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=3, n_hidden=32, n_layers=4) # X, Z and the speed s in
th = X / s + G * net(torch.cat([X, Z, s], 1)) # the balance, built in
thX = grad(th, X); thZ = grad(th, Z) # derivatives by autograd
r = (grad(thZ, Z) - 6*Z*(1 - Z)*G*s*thX) / G # residual, of order one
```

## FDM AND PINN, SCORED EXACTLY

|            | FDM            | PINN         |
|------------|----------------|--------------|
| as exact   | 401 × 41 nodes | 1.7e-04      |
| the field  | 3.4 ms         | 1.5 ms       |
| one number | 3.4 ms         | about 0.1 ms |
| new speed  | a new march    | read off     |
| offline    | none           | 2 minutes    |

Online, inference only, at the same accuracy (1 m/s), same device: about two times faster for the field, 30 to 40 for one number. The training is offline and reported apart; the speed as an input is the larger gain. Numbers: Ex\_09.2 section 5 on Colab's T4.

# Design by Gradient

**Design by gradient** — finding the parameter a design rule asks for from a PINN that has it as an input: the PINN gives the quantity, automatic differentiation its derivative with respect to the parameter, and Newton's method the value, with no further solve.

## NEWTON THROUGH THE PINN

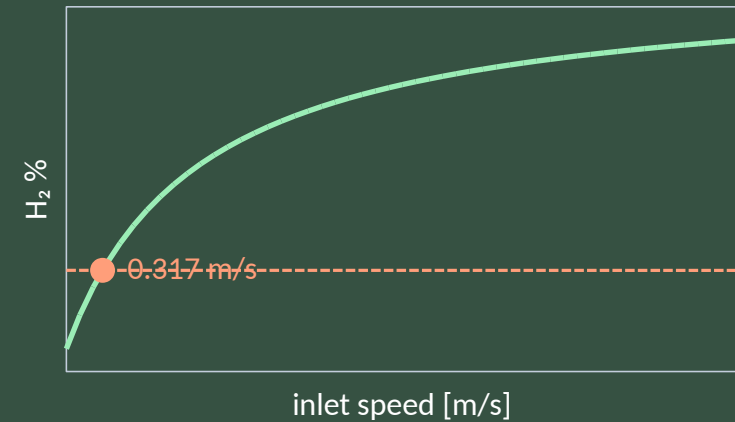
$$s_{k+1} = s_k - \frac{f(s_k) - f^*}{f'(s_k)}$$

$f(s)$  the hydrogen at the electrode at the outlet,  $f^* = 25\%$ ,  $f'$  by autograd

### THE DESIGN SPEED IN PYTORCH

```
s = (1 / theta_hand).requires_grad_()      # start: the hand calculation
for k in range(5):
    f = h2_at_electrode_outlet(net, s)      # the network, at X = 1, Z = 0
    s = s - (f - 0.25) / grad(f, s)        # Newton, the slope by autograd
```

## 25 % HYDROGEN AT THE OUTLET



|              |                        |
|--------------|------------------------|
| <b>PINN</b>  | 0.3173 m/s, five steps |
| <b>exact</b> | 0.3174 m/s             |
| <b>then</b>  | 74 % used, 8 Pa        |

# Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

| topic                        | what it adds                        | read             | where it makes a difference |
|------------------------------|-------------------------------------|------------------|-----------------------------|
| turbulence                   | eddies of every size                | White ch. 6      | every fast flow             |
| turbulence models            | the mean, and a model for the rest  | White ch. 6      | industrial CFD              |
| <u>Navier-Stokes problem</u> | can a smooth flow break down?       | White ch. 4      | the mathematics             |
| the Graetz entrance          | the first millimetre of the profile | Incropera ch. 8  | short channels              |
| several species              | each diffusing through the others   | Incropera ch. 14 | the air side, MP9.2B        |
| compressible gas             | density with the pressure           | White ch. 9      | Mach above 0.3              |

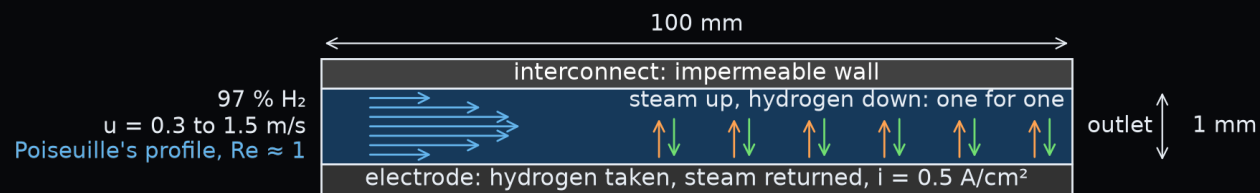
WHY THEY ARE LEFT OUT

The fuel channel's gas is slow, laminar and fully developed, with an exact solution to score against, so none of them is needed to do Ex\_09.2. The first three rows are turbulence, which no flow in this cell reaches.

## Exercise 9.2

### ONE NOTEBOOK, SECTION BY SECTION

- 1 – 3 — the fuel channel, its data, the exact field
- 4 — finite differences, marched from the inlet
- 5 — one PINN for every speed; timed at equal accuracy
- 6 — the inlet speed a design asks for
- 7 – 9 — the report; mini projects MP9.2A and MP9.2B



### THE QUESTION EACH PART ANSWERS

- 1 – 3 why is the gas simple?
- 4 how fine must the march be?
- 5 what does a PINN add?
- 6 which speed keeps 25 %?

### REPORT QUESTION, TO CONCLUDE

At the same accuracy the PINN answered two to forty times faster online, after two minutes of training, and found the design speed without a sweep. For which questions about this cell would you train a PINN, and for which march?

# Key Takeaways

## GAS AND SPECIES

|                    |                                |
|--------------------|--------------------------------|
| <b>what it is</b>  | a slow gas carrying a species  |
| <b>function</b>    | the steam fraction everywhere  |
| <b>train, tune</b> | exact: balance plus a quartic  |
| <b>vs a liquid</b> | the same flow; moles, not mass |
| <b>used for</b>    | fuel use, the electrode's gas  |

## ONE PINN, EVERY SPEED

|                    |                                  |
|--------------------|----------------------------------|
| <b>what it is</b>  | a PINN with the speed as input   |
| <b>function</b>    | the field at any speed, read off |
| <b>train, tune</b> | the balance built in; once       |
| <b>vs a march</b>  | faster online, no re-solve       |
| <b>used for</b>    | design by gradient               |

*Next — L10.1: the species equation again, inside a battery's particle.*

# Questions

## 1 - 2 · GAS AND SPECIES

### 1 Which three facts make the gas in a fuel channel a known, incompressible flow?

Its Mach number is below a thousandth, so its density hardly changes. The electrode returns one molecule for each it takes, so the moles per volume and the velocity stay the same. And the Reynolds number is 1.3, so Poiseuille's parabola holds from the inlet. *White ch. 9; Ex\_09.2 section 3*

### 2 Why can diffusion along the channel be dropped, and what does that make the equation?

The Péclet number along it is 125, so that term is a ten-thousandth of the one across. Without it the equation is parabolic in x: it marches from the inlet like heat in time and needs no outlet condition. A zero-slope outlet would be wrong, as the mean keeps rising. *Incropera ch. 14; Ex\_09.2 sections 3 - 4*

## 3 - 4 · ONE PINN, EVERY SPEED

### 3 What does the PINN's trial function build in, and why is the Sherwood number hard to get?

The balance X/s, the hand calculation, so the network learns only the profile across, in units of G. Both methods find the field to a ten-thousandth, but the Sherwood number only to within 8 to 31 %, because the profile it measures is a tenth of a per cent of the field. *Incropera ch. 8; Ex\_09.2 section 5*

### 4 What can a PINN with the speed as an input do that a march cannot?

Give the field at any speed without solving again, and its derivative with respect to the speed by autograd. Newton's method then finds the speed for 25 % hydrogen at the outlet, 0.3173 m/s against the exact 0.3174, with no sweep. At equal accuracy it also answers two to forty times faster. *Liu ch. 2.2; Ex\_09.2 section 6*

NEXT

## L10.1 — Ionic Diffusion

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The species equation again, inside the particle of a battery's electrode, where the measurement is a voltage at its surface.

### BEFORE THE NEXT SESSION

|             |                                     |
|-------------|-------------------------------------|
| read        | Incropera ch. 14                    |
| run         | Ex_09.2, sections 1 to 7            |
| bring       | your design speed, and its error    |
| think about | a parameter you would make an input |

*Block nine is finished: a liquid in a tube, and a gas that carries something along a channel.*